

Bi-encoders, Cross-encoders & CoBERT

BERT, sentence-transformers —
you've been using bi-encoders all along.

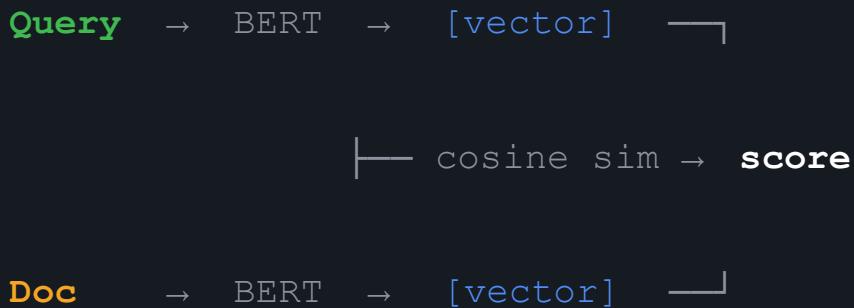
They work.

Until they quietly don't.

One vector. Whole document.

That's the bottleneck.

Whole doc → one vector. Specific details average out. Precision lost.



*Entire document = 1 point in space.
Specific detail = gone.*

Cross-encoder reads both together.

Quality jumps. Speed dies.

No precomputation. 100M docs = 100M BERT calls per query. Reranker only.

```
[ CLS ] query [ SEP ] doc [ SEP ]
```



BERT (full attention)

query ←→ doc tokens



score

Trade-offs

✓ **Accurate** ✗ Re-embeds per query ✗ Can't retrieve
✗ $O(N)$ forward passes

Keep every token. Interact late.

Late interaction: best of both worlds.

Docs encoded offline. Each query token finds its best doc token. Sum = score.

```
Query → BERT → [q1] [q2] [q3] ┐  
                                └─ MaxSim → score
```

```
Doc → BERT → [d1] [d2] ... [dm] ┘
```

MaxSim: each **qi** finds best **dj** → sum

Trade-offs

✓ Rich matching	✓ Docs offline	✗ Storage ×N	✗
Needs PLAID index			

None of these replaces the others. They stack.

Bi-encoder retrieves top 1000. ColBERT or cross-encoder reranks. Each does one job.

